

Bayesian-Inspired Inverse Problem for Advection-Diffusion Systems

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1 Abstract

This paper presents a stochastic inverse problem framework for estimating unknown parameters in an advection-diffusion equation. A Bayesian-inspired sampling method is used to recover velocity and diffusion coefficients from noisy observations.

2 Mathematical Model

The governing equation is:

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2}$$

where v is the advection velocity and D is the diffusion coefficient.

3 Methodology

We use a stochastic sampling approach:

- Random sampling of parameters (v, D)
- Numerical simulation of PDE
- Cost function based on L2 error
- Selection of best parameters minimizing cost

4 Results

True parameters:

- $v = 0.8$
- $D = 0.05$

Estimated parameters:

- $v \approx 0.7939$
- $D \approx 0.0497$

The method achieves less than 2% relative error.

5 Cost Function

The optimization minimizes:

$$J = ||u(v, D) - u_{true}||^2$$

Observed minimum cost:

$$J_{min} \approx 0.0415$$

6 Conclusion

The proposed method successfully solves the inverse problem with high accuracy and stable convergence.